Is LESS More? Applying Advances in Models of Human Behaviour to *Snake*

Adam Sigal
adam.sigal@mail.mcgill.ca
McGill University
Montreal, Quebec, Canada

ABSTRACT

In order for human-robot interaction to be possible, robots must be able to infer and predict human intent. To do this, they must employ a model of human behaviour. The current standard is the Boltzmann-noisily rational model of human behaviour [2, 5]. However, this model has shortcomings based on how it attributes probabilities in spaces that are either continuous or contain many similar options from which humans may choose. In their Award-winning 2020 paper, Bobu et al. propose a new human decision model called LESS (Limiting Errors due to Similar Selections) [4]. This model accounts for similarities of choices by weighting their probabilities accordingly. In this paper, the authors call for their model and findings to be tested in more complex environments. The present study applies the model to *Snake*, a ubiquitous grid-world style game. Within this context, LESS was employed on a human behaviour prediction task, and its performance was compared with that of the Boltzmann model. The model's applicability to this task was found to be limited, indicating a crucial failure case. This leaves the door open to further research into the effectiveness of LESS in more diverse environments.

KEYWORDS

ECSE681, roboethics, robotics, robot inference and prediction, human decision modeling

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1 INTRODUCTION

When humans interact with one another, they are able to estimate the actions and behaviour of others. This allows us to interact smoothly. Similarly, for robots to interact with us, they too must be able to estimate our behaviour. More specifically, they must be able to infer our intent through our actions, and anticipate our future behaviour to be able to successfully collaborate with us. The Boltzmann-noisily rational decision model has seen wide success in the modelling of human trajectory planning [2, 5]. It has been applied to fields such as mobile robotics [10], autonomous cars [11], and robotic manipulators [3] for both human inference [1] and prediction [11].

However, Boltzmann rationality is based upon Luce's axiom of choice, a decision model developed for the fields of econometrics and mathematical psychology [12, 13], and therefore has shortcomings when applied to robotics. A central assumption of Luce's axiom is that all options presented for a given decision are both discrete and *distinct*. However, human trajectory choices are both continuous, and on a "sliding scale" of similarity to one another. This produces a critical weakness of the Boltzmann model when applied to human trajectory inference and prediction.

The weakness created by the lack of a measure of *similarity* in the Boltzmann model was addressed in Bobu et al.'s paper: LESS is More: Rethinking Probabilistic Models of Human Behavior [4]. In it, the authors propose a new human behaviour model called **LESS** (Limiting Errors due to Similar Selections). They found that by weighting options according to their similarity, they were able to outperform the accuracy of the Boltzmann model in grid-world navigation tasks, robot inference through human simulation, performance under feature misspecification, and robot manipulation tasks. The authors stated that their findings were limited by the simplicity of the environments in which they were tested recommended that further study be conducted in more complex contexts.

The goal of the present study is to replicate the findings of Bobu et al.'s LESS paper, by comparing the two aforementioned human decision models (Boltzmann and LESS) on the game *Snake* [21]. While the game of snake is only incrementally more complex than the basic trajectory study presented in the original paper, there is a multitude of new dimensions of the problem that are added. Examples include human inconsistency due to factors such as diversity of grid configuration, human error due to misjudgement, fatigue, lack of skill, and speed of decision making. Furthermore, for gameplay to not be trivial, the grid space must be much larger than that presented in Bobu et al.'s study, which used a 7x5 grid for one of its inference studies. Potential limitations present in the original study such as feature misspecification are equally present in this experiment, and their effects have the potential to be scaled up along with the experiment's complexity. Permutations of these new dimensions of complexity can quickly lead to an exponentially larger scale of a given task, and stretch these behavioural models far beyond the domains of the experiments on which their results were demonstrated.

The application of these models to Snake will reveal how well each of them perform subject to these complications in terms of inference and prediction. Ultimately these models aim to be used

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to better parameterize human behaviour so that technologies can better interact with and serve people. The real world is an infinitely complex environment, so knowing how well LESS can be applied is crucial. This study aims to go one step further in exploring the validity of using these tools to model human behaviour in increasingly realistic settings.

2 BACKGROUND

Rudenko et al. published a survey in 2019 of the methods and approaches used in human motion trajectory modelling [15]. This survey can serve to provide context on Bobu et al.'s work as well as the current study. There are are three main approaches to modelling human motion prediction: physics-based methods, pattern-based methods, and planning based methods. Physics-based methods involve defining the dynamics of a given environment. This method is applied especially to define the probabilistic trajectory spaces of moving agents in safety-critical situations such as collision avoidance [6]. Pattern-based methods infer statistical behavioural patterns from trajectory data. Boltzmann and LESS, at least in the present application, fall under this category. More specifically they are *non-sequential* pattern-based methods, meaning that they do not consider a notion of time in their path considerations, although a version of this can be included in feature space, as in Bobu et al.'s robot arm inference experiment. Sequential pattern-based methods include architectures such as LSTMs and Transformers. Lastly, there are planning-based methods in which models explicitly reason about the agent's long-term intent. Examples of these methods often involve reinforcement learning or inverse reinforcement learning (IRL).

Non-sequential pattern-based models concern global motion patterns which are grouped together for analysis. This has the advantage of allowing for reasoning about behaviour on a higher level. It is often applied in a simplified or lower dimensional representation of a given environment, as opposed to physics-based models for example. This allows overarching patterns to more easily be interpreted.

As the present work is a replication study of Bobu et al.'s LESS model and its findings, it largely aims to contribute to the same goals. Behavioural models are the backbone of human robot-interaction. The goal of this work is to show how well the findings presented in the initial study can be applied to a wide variety of tasks, and do so with the conclusions of further work in the field taken into consideration. With LESS's widespread influence, it has already seen applications in published literature.

Wu et al. conducted a study proposing an inverse reinforcement learning algorithm that leverages the conclusions of Bobu et al. to better model the reward functions of human drivers [22]. The authors use continuous domain trajectory sampling on human drivers in both interactive and non-interactive scenarios, and were able to achieve more accurate prediction, convergence speed, and generalization than other comparable IRL algorithms. Approximation of the human's reward model is crucial in the performance of both Boltzmann and LESS. Without a truly representative reward approximation, these behavioural models lose meaning. As the task becomes more complex, the more unrealistic it becomes for humans to be able to accurately specify these reward functions by hand, as assumptions that must be made often become detrimental at larger scales. In the age of machine learning, this task is more often handed off to neural networks which model patterns in the data much more thoroughly than humans. However, limitations exist in machine learning methods, such as lack of generalizability, especially in situations where the amount of data available is small. Modeling reward functions using domain knowledge can be used to counteract such limitations, and Shah et al. argue that the best results are achieved with a mixed approach [17].

Furthermore, Giles et al. remark that human reward functions change over time, and propose that, not only must robots take this into account, but show that doing so allows robots to infer *more* than if they were simply learning a fixed behaviour model such as LESS [8].

Freedman et al. expounded on the effects of choice set misspecification [7]. The experiment conducted on this subject in Bobu et al.'s paper, while not as comprehensive, found that LESS was more robust to choice set misspecification than Boltzmann rationality. The authors classified the different types of misspecifications and found that some types had a neutral effect on future performance, while others often had acute negative effects. This work has potential to channel future work on minimization of model misspecification into more auspicious directions. The nature of feature misspecifications and their effects will also be discussed in this study. Ostensibly, to model a more complex setting sufficiently, more features are required, and the interaction between them must be more intricate.

The degree to which I was able to apply these conclusions in the present study were limited by a lack of experience with relevant data-driven approaches, time constraints, lack of access to computational power, and finally the need to keep the study within a scale appropriate for the final project of this course. I will discuss this further in Section 5.

2.1 The Game of Snake

Snake is a genre of game originating in the 1972 arcade game *Blockade* [9]. A version of the game was pre-loaded onto Nokia phones starting in 1998, and since then there have been countless iterations on various platforms. The game takes place in a grid-world environment. The player begins by controlling a "snake" that has a body that is a small number of blocks in length. When the player moves, the snake's body follows the path along which the player travelled. The player must navigate to a goal cell where there is "food" for the snake to eat. For every piece of food the snake eats, the length of its body increases by 1 cell. An *epoch* is defined as the moves taken by the player starting from the timestep wherein the snake reaches a cell containing food (or when the game begins), and the timestep wherein the player reaches the next food cell (or the end of the game). As the game progresses, the snake's body becomes increasingly long, and thus, harder for the player to avoid. The game ends when the snake's head collides with any part of its body. The player's score is equal to the number of pieces of food that the snake eats before a collision, or equivalently, the number of epochs played in a given game.

3 METHOD

The plan for this study was to take a set of games of snake played by a human, and from these data infer human preferences in trajectory planning in the context of the game. Once a model of this behaviour was inferred, performance of the Boltzmann and LESS models would be analysed on trajectory prediction tasks; determining how well the choices of each model reflect real human choices within the trajectory space. I will delve deeper into each of these steps in the following section.

3.1 Data processing and generation

3.1.1 Chosen Dataset. The initial plan for the study was to log 5-10 people playing Snake, and use this data for the experiment. However, following feedback received on this project's position paper, it was deemed to be sufficient to use data collected from a single player, and simply try and model the chosen individual's behaviour. In response I opted to use Michael Vine's Snake gameplay dataset, available publicly on kaggle.com [18].

This dataset contains data describing approximately 25 games played by the author on a 27x27 grid. Scores range from 0 to roughly 120, and the player is not an expert. After data cleaning, there were 2809 unique epochs (as described in Section 2.1). At each timestep, the following information is given to describe the state: a string representing the game's ID, the current score, the snake's length, the x and y coordinates of the snake's head, the current direction of travel, a boolean representing if the food (goal) cell is visible, the x and y coordinates the food cell, and a string representing a list of tuples of the coordinates on the grid occupied by the snake's body.

While it seemed as though using this pre-made dataset was going to save me time and effort, it was fraught with hidden issues, some of which were not easily noticed until late stages of the study. This issue was exacerbated when I realized that since this dataset did not use a common random seed allowing for a smaller set of snake-food configurations, comparison between games was more difficult. Instead of having multiple examples of player behaviour on a given configuration, as was studied in the grid-world experiments of Bobu et al., each configuration had only one path taken by the player. This made inference of player preference from each A-to-B trajectory less meaningful, as patterns could not be gleaned from multiple passes through the same state. In a situation such as this, a more intricate reward function must be used, and more data must be present in order to reasonably model behaviour. This presented a central limitation of the results that could be gathered in this study, as will be addressed further in Section 5.

3.1.2 Data generation. To do prediction using the Boltzmann and LESS models, there has to be a set of options to evaluate and choose from. Since there was only one path taken by the player at each epoch, other options had to be presented in order for prediction to be possible. The trajectory space of a given Snake configuration varies. In most configurations where the snake's body is not too long, the space is actually infinite. For example the player could spin in a circle for an arbitrarily long time before deciding to go get the food. Even in more limited settings, the trajectory space is exceptionally large, especially on a 27x27 grid. It is therefore necessary to sample the trajectory space and use these sets for the prediction task. Note that inference is performed only on human paths, so the generation method does not influence the inferred model of human behaviour.

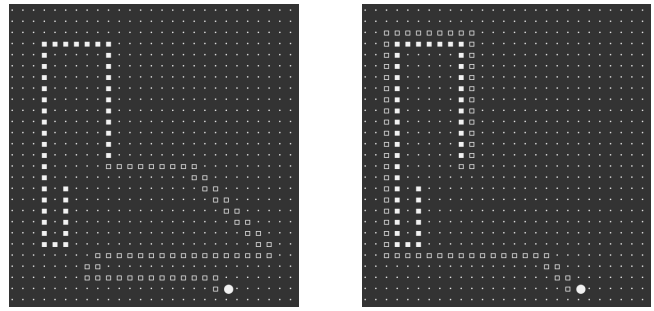


Figure 1: Pseudo-randomly generated trajectories for a single initial configuration (solid boxes represent current snake location; empty boxes represent generated trajectory

The task was now to determine how to best generate trajectories for the current context. A task for which I did not have any domain experience from which to draw. In researching the topic I found only methods that were either not applicable or overly complex. What I gathered from this research was that the best option would be to find the optimal path from A to B using a pathfinding algorithm (I chose A*), and perturb it somehow in order to generate different options [19].

The method I settled on is to generate *waypoints* to which the snake would successively travel using the A* algorithm. The number of waypoints was generated using a Gaussian distribution as described in (1):

$$n \sim \mathcal{N}(1.5; 0.6)\,, \text{ where } n \geq 1 \tag{1}$$

Where n is the number waypoints. This equation is designed such that there are most commonly two waypoints, followed closely by having one waypoint, and decreasing in likelihood from there. There is always at least one waypoint, because otherwise the generated path would simply be the optimal one, and we do not want exact repetitions in the sample set. The coordinates of each waypoint are chosen using (2).

$$x, y \sim \mathcal{U}(1, 27)\,, \quad \text{where } (x, y) \text{ is empty} \tag{2}$$

If it is possible to travel between the n waypoints, then the path is added to the sample set.

In total, 100 paths were generated for each of the 2809 epochs, giving a total of 280,900 paths generated. Examples are shown in Fig. 1. As in Bobu et al.'s study, the next step was to define the features used to describe each path.

3.2 Inference, Feature Specification and Reward Function

3.2.1 Feature Specification. Bobu et al. defined feature space differently for their different experiments. In one experiment the authors only studied left-right bias in equivalent trajectories, so trajectories were described using a one-dimensional binary feature space: $\phi(\xi) \in \{Left, Right\}$, where ξ represents a trajectory, and $\phi(\xi)$ represents its feature vector. Later experiments use various feature

spaces including distance from obstacles, trajectory length, average trajectory coordinates, and / or trajectory efficiency as measured by speed.

The parameters I decided to track were: the length of the snake's body (also equal to current score), the length of the path taken to the food, the degree of closeness to the snake's tail (obstacle distance), the optimality of the path, and the distance to the goal. However, since goal distance and snake length are constant for any path taken from a given configuration these were not included in the final feature space. I hypothesize that these are, in fact, useful features to describe the nature of a trajectory, and could be useful in modeling human behaviour depending on the configuration. It is likely these would influence how the other features are weighted in a hypothetical human reward function. However, since the reward function used in this study is a simple dot-product between weights and corresponding features, this interaction could not be captured, and thus these features were not included.

Therefore, trajectories were modeled using 3 features path length l, optimality o, and obstacle distance d:

$$\phi_{raw}(\xi) = (l, o, d), \quad \text{where} \;\; l \in \mathbb{N};\; o, d \in \mathbb{R}^{+} \tag{3}$$

Optimality and obstacle distance were calculated using the symmetric Hausdorff distance, a measure of distance between two sets [16, 20]. Optimality is equal to the symmetric Hausdorff distance between the given trajectory, where the optimal trajectory as determined using the A* algorithm. Obstacle distance is equal to the symmetric Hausdorff distance between the given trajectory and the snake's body at the initial timestep of the corresponding epoch.

Due to computational considerations, it was important to keep the weighting of the features in a small range, so that it could be discretized and iterated through in later calculations. For weighting within such a small range to be meaningful, the feature space had to be normalized. This was carried out on the full set of player data. For each feature, its mean was subtracted (4), and it was divided by its variance (5).

$$\phi'(\xi) = \phi_{raw}(\xi) - \mu_{\phi_{raw}} \tag{4}$$

$$\phi = \frac{\phi'(\xi)}{\sigma^2_{\phi'}} \tag{5}$$

Where $\mu_{\phi_{raw}}$ is the mean of ϕ_{raw} and $\sigma^2_{\phi'}$ is the variance of ϕ'. With this normalization, $\phi(\xi) \in \mathbb{R}^3$

3.2.2 The Reward Function. The Boltzmann and LESS models operate on the assumption that behaviour is dictated by humans approximately optimizing an intrinsic reward function. Bobu et al. do not explicitly describe the reward function(s) that were applied in their experiments, so I had to design one that I felt to be appropriate for the present context. The design of a highly accurate reward function is an entirely different problem than is being addressed in this work. Since I both needed to keep the scope of this project focused on the two decision models, as well as not having experience in designing reward functions (nor the time to learn enough), I opted for the simple equation in (4).

$$R(\xi, \theta) = \sum_i \theta_i \cdot \phi_i(\xi) \tag{6}$$

This formula assigns a single weight (or parameter) to each element of a trajectory's feature vector, and takes the sum thereof. The weights are contained in the vector θ and each of its elements are learned through inference on the player's data. Each weight θ_i takes a value in the range $[-1, 1]$ which is deemed sufficient because the feature space has been normalized as described in section 3.2.1.

3.2.3 Inference. As in Bobu et al.'s paper, inference is performed using trajectory sample sets of different sizes: 10, 30, 100, 300, and 1000, as well as 2809 (full dataset). The authors found that LESS performed better on small sample sizes, while Boltzmann did increasingly well with sample size, eventually performing slightly better due to overweighting of uncommon examples. I expected to see similar results on the current game data. However, since in this study I did not carry out separate inference and prediction tasks due to time and computational constraints, results will not exclusively reflect inference performance. This is a limitation of the study and will be discussed further in Section 5.

$$b'(\theta) = \frac{b(\theta)p(\xi|\theta)}{\int_{\Theta} b(\bar{\theta})p(\xi|\bar{\theta})d\bar{\theta}} \tag{7}$$

The initial paper presents equation (5) for Bayesian inference using LESS. While it is seemingly a simple implementation of a Bayesian update, multiple problems arose when trying to implement it. Firstly, as discussed by Bobu et al. implementing this integral is essentially intractable. Even with a reasonable discretization of the parameter space, computing the desirability of every potential weight vector on a meaningfully sized trajectory sample set was deemed to be too computationally expensive. For each potential θ, the likelihood of the given trajectory would have to be calculated, as well as the sum of the likelihoods of the trajectory given every other $\bar{\theta}$ option (note that the bar on $\bar{\theta}$ is used as a distinction marker as in the original paper, and does not represent mean). This process would be repeated for the number of trajectories upon which inference was carried out. I will discuss my decision, as well as its implications further in Section 5, but on the recommendation of a student tutor from the McGill statistics help group (Statistics Online for Students (SOS)), I opted for a simpler parameter inference method [14].

In this method, the weighting θ^*_{ξ} that maximizes the reward $R(\xi, \theta)$ is calculated for every trajectory ξ in the sample set as in (8). The overall inferred θ^* is calculated to be the average of each of these weights as in (9):

$$\theta^*_{\xi} = \underset{\theta}{argmax}\; R(\xi, \theta) \tag{8}$$

$$\theta^* = \frac{1}{|\Xi_f|} \sum_{\xi \in \Xi_f} \theta^*_{\xi} \tag{9}$$

Where Ξ_f is the finite sample set of trajectories.

Unfortunately, the decision to use this method removes the study of Boltzmann and LESS inference from this work. I was dissatisfied with this, but under the time constraints I had no other option. However, one good consequence of these circumstances is that the final results will exclusively reflect the prediction performance of the decision models, allowing for more direct comparison, Still, if

the parameterization θ^* will not as accurately reflect player preferences in the reward function, it will hinder the Boltzmann and LESS models' ability to draw patterns from the data and predict behaviour.

3.3 Prediction: Boltzmann and LESS Models

Armed with the calculations shown in Section 3.2, it was now possible to calculate decision probabilities according to the Boltzmann (10) and LESS (11) Models as described in Bobu et al.'s initial paper. The authors present a discrete version of LESS (14) but due to ambiguities that I did not have the time or resources to ascertain, I opted to use the result in (11).

$$p(\xi) = \frac{e^{R(\phi(\xi))}}{\sum_{\bar{\xi}\in\Xi} e^{R(\phi(\bar{\xi}))}} \tag{10}$$

$$p(\xi) \propto \frac{e^{R(\phi(\xi))}}{\int_{\Xi} s(\phi(\xi),\bar{\xi})\, d\xi} \approx \frac{e^{R(\phi(\xi))}}{\sum_{\Xi_f} s(\phi(\xi),\bar{\xi})} \tag{11}$$

Where $s(x,\xi)$ is the similarity function as given by (13):

$$s(x,\xi) = \left(\frac{1}{\sigma\sqrt{2\pi}}\right) exp\left(-\frac{\|x-\phi(\xi)\|^2}{2\sigma^2}\right) \tag{12}$$

which, in turn, is paramaterized by the bandwidth σ. The expression to calculate the optimal bandwidth σ^* was given by (13).

$$\sigma^* = \underset{\sigma\in\mathbb{R}}{argmax} \sum_{\xi\in\Xi_f} log\left(\sum_{\bar{\xi}\in\Xi_f} s(\phi(\xi),\bar{\xi})\right) \tag{13}$$

Unfortunately, when carrying out the process described in (13), σ was found to follow a logarithmic form (Fig. 2). That is to say, the process did not work analytically. It does not seem as though I implemented it incorrectly or misunderstood anything, but I cannot say with certainty that this result is not due to human error on my part. Since no optimal σ^* could be determined, it was chosen to be $\sigma^* = 1$.

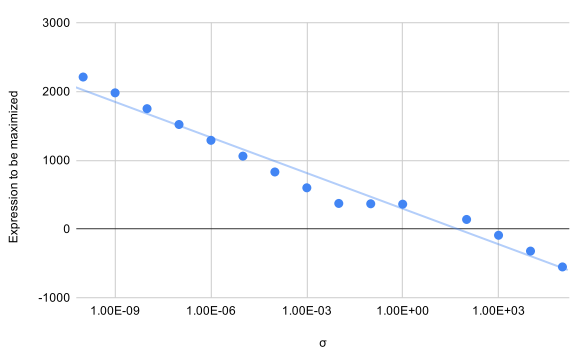


Figure 2: Analytical process to find optimal bandwidth σ^*

3.3.1 Performance analysis. Bobu et al. used metrics such as *True-Posterior*, aggregate Kullback-Leibler divergence, and Turkey HSD, among others, to analyse their results. I had originally planned to use as many of the same metrics that I could in order to have the most comparable results possible. Unfortunately, after researching many of these methods, I found that I did not have the background necessary to learn how to properly use these tools within the time frame of this project. I instead was obligated to do a more naive analysis, wherein I compared selection probability under Boltzmann and LESS against similarity to player trajectory. Theoretically, more similar generated paths should be more likely to be chosen.

It would have been interesting to perform k-means clustering on this data to see if different clusters of data exhibit any unique patterns or features, but I was not able to familiarise myself with this method within the time frame of this project. Similarly due to time constraints, I was only able to analyse the results on the first of the games for which data was collected, containing 54 unique initial state configurations. This is far from optimal, but as will be revealed in the Results section, this smaller amount of data was ample enough to see the trend of the results.

4 RESULTS

The similarity to player path of generated paths was found to follow an exponential distribution (Fig. 3). This corresponds with expectations as paths are sampled from the trajectory space uniformly (1), (2), while the similarity function (12) follows an exponential distribution. However the maximum value of similarity was empirically found to be 0.4, though it should range between 0 and 1. This is likely a symptom of the failure to calculate the proper optimal bandwidth as described in (13). This miscalibration of the similarity function propagates forward to LESS, where ultimately it hindered the model's performance in the present prediction task.

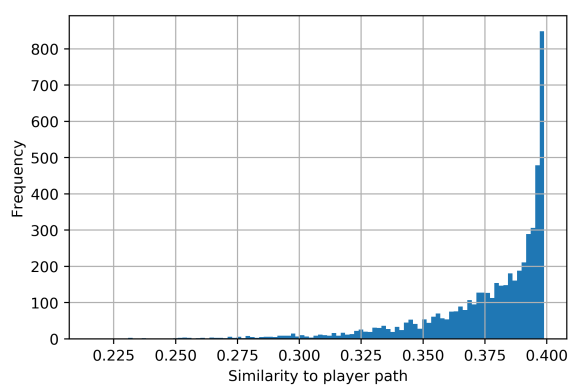
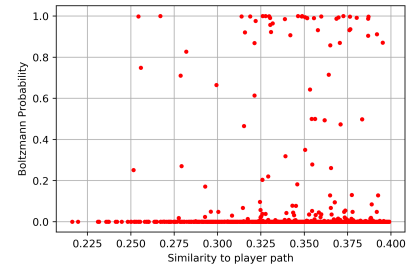
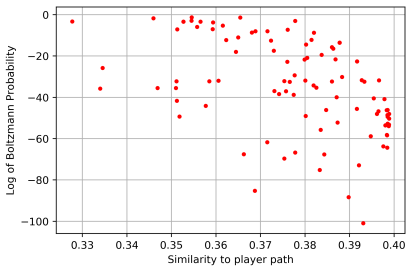


Figure 3: Histogram of pseudo-randomly generated trajectories' similarity to corresponding player trajectory

A first glance at Fig. 4 could lead to believe that there is a positive correlation between similarity to the player path and the Boltzmann probability of selection. This would have indicated that inference on the weighting of trajectory features was successful such that the model learned player preferences and selected paths from the sample set that more closely met those criteria. Unfortunately, as shown

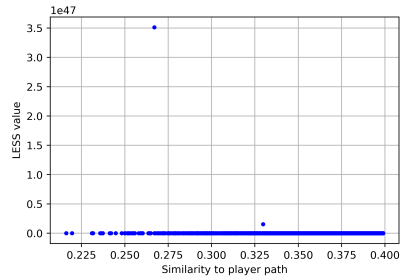


(a)

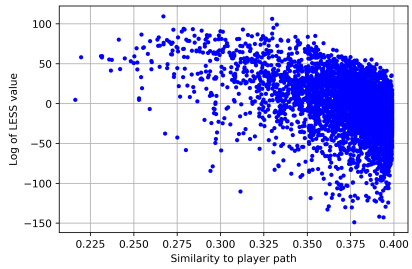


(b)

Figure 4: (a) Boltzmann probability as a function of similarity to player path, (b) log of Boltzmann probability as a function of similarity to player path



(a)



(b)

Figure 5: (a) LESS value as a function of similarity to player path, (b) log of LESS value as a function of similarity to player path

by the data presented in Fig. 3, random paths with high similarity to the player path are simply more frequent. Thus, we can reason that most of these high-Boltzmann-probability datapoints, are simply outliers, occuring at the same rate across all randomly generated paths. In fact, in Fig. 4b, we observe that the highest concentration of the generated trajectories cluster around the $log(p(\xi)) = -50$ point, which is equivalent to $p(\xi) \approx 2 \times 10^{-}22$ This illustrates shortcomings in the reward function as well as the inference of its parameter vector θ. In this data we see frequent dominance of the probability distribution by one or a small number of trajectories in the sample set, independent of similarity to the player path. This ultimately indicates that the use of the Boltzmann-noisily rational model on the prediction of trajectory selection was not successful.

What is most salient when observing the data in Fig. 5a is that the value produced, in fact, is *not* a probability, rising into the order of 10^{47}. Indeed, in equation (11), the expression is *proportional* to the probability of choosing the trajectory, not equal. I even tested simply dividing each of these LESS values by their collective sum, which in turn produced the exact same probabilities as produced by Boltzmann prediction. I will discuss this issue further in Section 5.

Taking this data as it is, there is a slight trend for lower LESS selection probabilities in the high-frequency area of high player path similarity, as seen in Fig. 5b. This would support the findings and goal of the LESS model, as trajectories in a cluster of similarity are supposed to be de-weighted in order to have them considered as a group. However, this trend is visible specifically when the logarithm of the LESS value is plotted, which is not something that Bobu et al. mentioned in their initial paper. Ultimately I did not perform an analysis of the statistical significance of this trend, and I hypothesize that it would not meet the threshold, due to the numerous problems in the data, reward function, and parameterization of the LESS function itself.

It would have been interesting to perform a multi-modal analysis on the Boltzmann and LESS prediction data, such as k-means clustering in order to the analyse patterns therein. Unfortunately, I did not have the time to refresh my knowledge on such methods to a point at which I would have been able to conduct a meaningful analysis. Nevertheless, after analysing the above data, I hypothesize that this method would have demonstrated similarly inconclusive results. This conclusion also applies to the other analysis methods used in Bobu et al.'s paper, as mentioned in Section 3.3.1.

5 DISCUSSION

5.1 Experimental Design and Method

5.1.1 Experimental Design. One confounding factor present in the initial experimental design was that the performance of both inference and prediction tasks were both going to be reflected in the final results. Ultimately, the Boltzmann and LESS inference component ended up being removed from the study as discussed in Section 3.2.3. However, the effects of low-quality inference still persisted in the final results, and were a major contributing factor in causing the results to be inconclusive.

In designing this experiment I did not anticipate my degree of knowledge in probability concepts and statistical analysis to be a

limiting factor, but it was. It was not until attempting to implement many of the equations in Bobu et al.'s paper that I realized that there were gaps in my experience that prevented me from progressing. I had issues implementing the equations for Bayesian inference, LESS, and optimal bandwidth calculations. I believe that if I was more comfortable with the relevant probability concepts, I may have been able to better troubleshoot these issues. Furthermore, I made the flagrant error assuming that I would be able to easily learn to use the statistical analysis methods that the authors used. Researching these methods with the intent of implementing them, I quickly realized that I did not have the necessary background to learn to use them correctly within the given time frame.

5.1.2 Data Processing. As discussed in Section 3.1.1, the fact that the dataset that I chose did not have repetitions of initial state configurations meant that it was more difficult to infer patterns in human preferences. This would not have been the case had I implemented the game myself in such a way that multiple player paths could be collected for a single configuration. In fact, if there had been multiple player paths in the data for every unique configuration, my naive inference method may have been more likely to extract meaningful results. That being said, this was not the most consequential issue present in the dataset.

There were inconsistencies with the dataset that I selected. The author reported that the game was played on a 26x26 grid. This was not something I thought to question, and was not obviously wrong from skimming the contents of the dataset. For this reason, it was not until late in the data engineering process that I noticed that the data was, in fact, collected on a 27x27 grid; coordinates were zero-indexed, so the *maximum value* was 26. Secondly, there were inconsistencies in game identification within the dataset. The online repository provides two versions of the dataset: one with a list of coordinates of the snake's entire body at every step, and one without. These two versions not only use different game identification numbers, but split the data at different points. Furthermore, the version with the snake coordinates had multiple games grouped under a single identification number, and both contained duplicate games. These issues made it difficult to comprehend the true nature of the dataset, but by the time all of these issues had accumulated, too much time had been invested into it to consider alternatives.

5.1.3 Trajectory generation. Not having much experience with the subject, the method I chose for trajectory generation was quite naive. As I described in Section 3.1.2, I did research into the field's common practices, however the specifics of what I found were often outside of the scope of my background, and thus I could only extract simple suggestions. In the current implementation of Snake, the trajectory space from a given initial state is infinite most of the time. Sampling an infinite space with minimal heuristics inevitably leads to a preponderance of low quality datapoints being included in the sample set. This is especially true given that due to computational constraints, only 100 trajectories were sampled for each initial state. For the Boltzmann and LESS models to produce interesting results and find meaningful patterns, there must be a diverse and reasonable set of trajectories from which to choose. Since this was not the case in this study, the significance of the future results were diminished.

For the above two tasks, the major limiting factor were my skills at processing data. I overestimated my personal capacities as well as underestimating the difficulty of what needed to be done. Because of this, the data engineering process took an inordinate portion of the project's timeline. Once the data was ready to use in inference and prediction tasks, I had much less time than was truly needed to focus on the principal parts of the study. When I similarly ran into issues during those portions of the project, I simply did not have enough time to manage and resolve them properly.

5.2 Inference, Feature Specification and Reward Function

5.2.1 Feature Specification and Reward Function. Like I touched on in Section 3.2.1, there were many features that I considered for modelling trajectories. Important ones that I think are crucial in order to truly approximate a human reward function are current snake length and goal distance. Behaviour is observably different depending on these factors. However, the simple linear reward function that was chosen for this work could not model interactions between the different parameters in a feature vector. I believe it would be more appropriate, for example, to use a small neural network so that truly representative relationships can be learned between these parameters, rather than a simple weighting of each one. There are likely further appropriate reward function considerations applicable to this context, but the topic of reward function design is vast, and thus I deemed it unrealistic to attempt become sufficiently educated on the topic for the scope of this project. That being said, both the Boltzmann and LESS models hinge on the choice of reward function, and the trivial option that was chosen severely limited their eventual performance.

5.2.2 Inference. As it has been years since my pure probability classes, I found that I was less comfortable with some of the concepts than I had anticipated. This was an acute issue in trying to implement inference via Bayesian update as described in equation (7). While my initial assessment that it would have been very computationally expensive to implement was likely correct, I now think that I could have reasonably done it. Whether I could have done it in the timeframe within which I was working is less apparent. When consulting with the McGill Statistics peer tutor, he suggested that the belief function in this equation has to be defined by the experiment designer. This was what finally convinced me to abandon using Bayesian inference in this study, since Bobu et al. made no mention of the belief function used. Looking back, I believe that the original authors intended it to represent a prior probability, but since I am out of practice on these concepts, I may be incorrect. I believe that my naive inference method described in (8), (9) gave lower-quality results than what Bayesian inference with Boltzmann and LESS would have, since it did not make probabilistic considerations. To make matters worse, since the player was often very sub-optimal and subject to mistakes and human error, this made inference even more difficult in light of the simple feature space, reward function, and rather small dataset size.

5.3 Prediction: Boltzmann and LESS Models

The implementation of the Boltzmann model was straightforward and did not cause issues. As previously discussed, there were many

previous issues that propagated forward and contributed to its inconclusive results on the prediction task. This is also true of LESS, but with added issues in its implementation. Firstly, there were inconsistencies and vague language surrounding the description of the model's formula in Bobu et al.'s paper. Much of this was due to my lack of familiarity with the field, but I assert that some aspects were truly inconsistent or insufficiently explained. For example, there were ambiguous substitutions of $\phi(\xi)$ and ϕ in some instances, most notably in the discrete version of the LESS formula (14).

$$p(\xi) = \frac{e^{R(\phi(\xi))}}{\sum_{\bar{\phi} \in \Phi_{\Xi_f}} e^{R(\bar{\phi})}} \cdot \frac{s(\phi(\xi), \xi)}{\sum_{\bar{\xi} \in \Xi_f} s(\phi(\xi), \bar{\xi})} \tag{14}$$

Although the authors touch on this, I still was not able to determine with certainty what they meant to convey by both parameterizing the reward function with $\phi(\xi)$ and ϕ in the same equation. They suggested that in the sum in the denominator of the first term, ϕ is meant to represent individual features from the feature space, but were unclear as to if we are to work in full feature space or one-by-one for the rest of the formula. If it is in fact one feature at a time, this poses issues because the reward function is not necessarily defined to be used with varying dimensions, nor is the similarity function. Furthermore, in the numerator of the second term, the similarity function, when given the arguments $s(\phi(\xi), \xi)$, always evaluates to 1, so I do not understand its utility. They touch on the fact that this evaluates to 1 under such circumstances, but do not mention that this *always* happens, leading me to believe that there is a gap in my understanding. Ultimately, when I encountered this problem, I did not have the resources or time to seek sufficient help on the subject. I instead opted to use the version of LESS described in (11). However, as seen in the Results section, the values calculated using this formula gave doubtful results, ranging into the order of 10^{47}, while supposedly being directly proportional to probability values. Seeing these results, and with my difficulty in comprehension in mind, I assume that there are notions in these calculations that I was unable to grasp. These issues demonstrate that I had overestimated my understanding of these formulas when I had initially planned this study, and that was a fatal misstep.

6 CONCLUSION

This study aimed to reproduce Bobu et al.'s paper introducing the LESS model of human behaviour on the game of Snake. LESS proposed the addition of a similarity metric as a solution to the weakness of the Boltzmann noisily-rational model in feature-dense areas of a trajectory space. LESS was a paper with widespread influence, and thus it is still important to verify its findings in diverse settings. The present study failed to verify or discredit the findings of the initial paper for a multitude of reasons. I found that I had not been thorough enough in my experimental design process, and that I had overestimated my abilities in data engineering, probability, and statistical analysis. Despite this, from this work I contribute a cleaner, more robust dataset based on Snake gameplay, as well as tools to visualise and process the data. This work was not a success, but the the applications of LESS and study of modelling human behaviour as it applies to human-robot interaction is still promising, and there lies great potential in its exploration.

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